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Math 215A

Homework 2
Due: October 7, 2026

Rings are understood to be commutative, unless stated otherwise.

For the sake of notation, let

$$[a_0, ..., a_n] = \sum_{k=0}^{n} a_k x^k \in A[x]$$

and $a_n \neq 0$.

Problem 1

Let k be a field. Show that $k[x]$ and $k[x, x^{-1}]$ are not isomorphic as k-algebras. (Here $k[x, x^{-1}]$ can be defined as the ring of Laurent polynomials $a_{-n}x^{-n} + ... + a_n x^n$, where $n \geq 0$ and $a_i \in k$.)

Solution. Assume the contrary, that $k[y]$ and $k[x, x^{-1}]$ are isomorphic as k-algebras. Thus there exists a $L : k[x, x^{-1}] \to k[y]$ a K algebra isomorphism. Then $L(x) = f = [a_0, ..., a_n]$ is a unit because x is a unit in $k[x, x^{-1}]$. Thus exists $g = f^{-1} \in k[x]$.

$$g := [b_0, ..., b_m].$$

$$1 = fg = a_0 b_0 + ... + a_n b_m y^{mn} \Rightarrow a_n b_m = 0.$$

But $a_n \neq 0 \neq b_m$. A field has no nonzero zero divisors. Thus a contradiction occurs.

Problem 2

Let k be a field. Show how to view the ring $k[x, y]/(x - 1, x^2 + y^2 - 1)$ as the quotient ring of $k[y]$ by a certain ideal. Deduce that the ideal $(x - 1, x^2 + y^2 - 1)$ is not prime, and compute its radical.

Solution. Note that

$$\frac{k[x, y]}{(x - 1, x^2 + y^2 - 1)} \cong \left(\frac{k[x, y]}{(x - 1)}\right)/(x^2 + y^2 - 1)$$

and the following isomorphism:

$$\frac{k[x, y]}{(x - 1)} \cong k[y]$$
$$1 \to 1$$
$$x \to 1$$
$$y \to y$$

Under this isomorphism, we have $(x^2 + y^2 - 1) \to (y^2)$. Thus

$$\frac{k[x, y]}{(x - 1, x^2 + y^2 - 1)} \cong \frac{k[y]}{(y^2)}.$$

> $y * y = 0 \in k[y]/(y^2)$, thus the ring isn't a domain. Hence the ideal isn't prime.

Problem 3 [TODO]

> (a) Let R be a domain. Show that the polynomial ring $R[x]$ is a domain and that the group of units $R[x]^*$ is equal to R^* (viewed as constant polynomials). Give an example showing that this description of $R[x]^*$ fails for R not a domain, and say where your proof fails. By induction on n, it follows that the polynomial ring $A = k[x_1, ..., x_n]$ over a field k is a domain, and that $A^* = k^*$. [TODO]

(b) Show that the power series ring $B = k[[x_1, ..., x_n]]$ over a field is also a domain, and find the group of units B^*.

Proof: **(a)**: Note that trivially, $R^\times \subseteq (R[x])^\times$. We have to show the opposite containment. Assume the contrary, that $f \in (R[x])^\times$ and $\deg(f) = n \geq 1$, thus $f = [a_0, ..., a_n]$ where $a_n \neq 0$. There exists $g = [b_0, ..., b_m]$ a multiplicative inverse of f.

$$1 = fg = a_0b_0 + ... + a_mb_nx^{m+n} \Rightarrow a_mb_n = 0.$$

However since both a_m and b_n are nonzero and R is a domain, they cannot multiply to 0. Hence a contradiction occurs. Thus the only units in $(R[x])^\times$ are 0 degree polynomial (constant polynomials) and to be a unit they must be in $R^\times$. Note that $R[x]$ is a domain as well (look at highest degree in any multiplication of polynomials). Now use the isomorphism $R[x_1, ..., x_n] \cong R[x_1, ..., x_{n-1}][x_n]$ together with the inductive hypothesis, $(R[x_1, ..., x_{n-1}])^\times \cong R^\times$ and the above argument to get the more general statement.

The issue that occurs when R is not a domain, is that our argument relies on the fact that there are no nonzero zero divisors ($a_mb_n = 0$) which is not true if R is not a domain. Let $R = \mathbb{Z}_4$, 2 is a zero divisor.

$$(1 + 2x)^2 = 1 + 4x + 4x^2 = 1 \in \mathbb{Z}_4[x].$$

Thus $(1 + 2x) \in (\mathbb{Z}_4[x])^\times$

(b): Let $B_n := k[x_1, ..., x_n]$ and $B_0 = k$. Note the canonical isomorphism, $B_n \cong B_{n-1}[[y]]$. Let $M_n : B_n \to \mathbb{Z}_{\geq 0}$ where $f \in B_n$ goes to the minimal $a + 1$ where $P_a \neq 0$ where

$$f = \sum_{k \geq 0} P_k y^k$$

where f is viewed as a element of $B_{n-1}[y]$. If no such a exists (ie $f = 0$), then $M_n(0) = 0$.

For $f \in B_n$ to be a unit a necessary condition is that the constant term must be a unit (we will prove that its a sufficient condition). B_0 is a domain, $B_0^\times = K^\times$. Suppose B_k is a domain that

$$(B_k)^\times = \{f \in B_k : \text{constant term of } \ f \text{ is } 0\}$$

Suppose $f, g \in B_{k+1} \cong B_k[[y]]$ nonzero thus $M_k(f), M_g(f) \geq 1$. Let $a := M_k(f) - 1$ and $b := M_k(g) - 1$.

$$f \coloneqq \sum_{k \geq a} P_k y^k$$
$$g = \sum_{k \geq b} Q_k y^k$$

where $P_k, Q_k \in B_k$. Thus

$$fg \coloneqq P_a Q_b y^{a+b} + O\big(y^{a+b+1}\big).$$

Since B_k is a domain, $P_a Q_b \neq 0$ this means that $fg \neq 0$. Hence $B_{k+1} = k[[x_1, ..., x_{k+1}]]$ is a domain. Suppose $f \in B_k[[y]] \cong B_{k+1}$,

$$f \coloneqq \sum_{k \geq 0} P_k y^k$$

where $P_0 \in (B_k)^\times$. We will define a inverse for f whose coefficients are recursively defined. Let

$$Q_0 \coloneqq \frac{1}{P_0}$$
$$Q_{k+1} = -\frac{1}{P_0} \sum_{j=1}^{k+1} P_j Q_{k+1-j}$$

Let

$$g \coloneqq \sum_{k \geq 0} Q_k y^k.$$

Now

$$fg = \sum_{k=0}^{\infty} \left(\sum_{l=0}^{k} P_k Q_{k-l} \right) x^k.$$

Looking coefficient by coefficient for $k \geq 1$ we see:

$$\sum_{l=0}^{k-1} P_k Q_{k-l} + P_0 \left(-\frac{1}{P_0} \sum_{l=0}^{k} P_k Q_{k+l} \right) = 0$$

For $k = 0$, we see $P_0 Q_0 = 1$. Thus $fg = 1$. Thus

$$\begin{aligned} B_{k+1}^\times &\coloneqq \left\{ [P_n]_{n \in \mathbb{Z}_{\geq 0}} \in B_k[y] : P_0 \in B_k^\times \right\} \\ &= \left\{ [P_n]_{n=0}^{\infty} \in B_k[y] : \text{constant term of } P_0 \in k^\times \right\} \\ &= \left\{ f \in B_{k+1} : \text{constant term of } f \in k^\times \right\} \end{aligned}$$

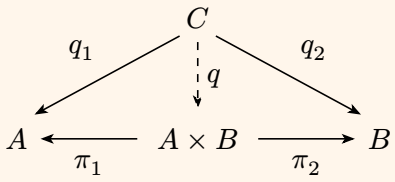
□

Problem 4

Let A and B be commutative rings. The product ring $A \times B$ (not to be confused with a tensor product) is the product set, with ring structure $(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 + b_2)$ and $(a_1, b_1)(a_2, b_2) = (a_1 a_2, b_1 b_2)$. State and prove a universal property that characterizes $A \times B$ in the category of commutative rings. (If this is not familiar, see the section on category theory in chapter 1 of Serge Lang's book *Algebra*.)

Proof: In the category of commutative rings, the product ring has the universal property of the product. □

Proposition 0.1. $A \times B$ exhibits the universal property of the product. In other words, if there exists a commutative ring C with ring homomorphisms $q_1 : C \to A$ and $q_2 : C \to B$ there exists a unique ring homomorphisms $q : C \to A \times B$ such that the resulting diagram commutes.



where π_i is projection map.

Proof: We will first prove existence and then uniqueness of q. Let $q : C \to A \times B$ where $q(c) = (q_1(c), q_2(c))$. This is a ring homomorphism, because q_1, q_2 are ring homomorphisms. Now for uniqueness, suppose $q : C \to A \times B$ and $q' : C \to A \times B$ cause the diagram to commute. We know that

$$(\pi_1 \circ q)(c) = (\pi_1 \circ q')(c) \ = q_1(c)$$
$$(\pi_2 \circ q)(c) = (\pi_2 \circ q')(c) = q_2(c).$$

Thus $q(c) = (q_1(c), q_2(c)) = q'(c)$. Hence $q = q'$. Hence q is unique. □

Problem 5

For rings A and B, give an example to show that an (additive) subgroup of the ring $A \times B$ need not be of the form $S \times T$, for subgroups $S \subset A$ and $T \subset B$. (The question is about equality of subgroups, not isomorphism of subgroups.) But show that every ideal in $A \times B$ is of the form $I \times J$ for ideals $I \subset A$ and $J \subset B$. Building on that, show that $\mathrm{Spec}(A \times B)$ is the disjoint union of $\mathrm{Spec}(A)$ and $\mathrm{Spec}(B)$, as a set. (In fact, it is the disjoint union as a topological space, but you need not prove that.)

Proof: Consider $\mathbb{Z}^2$ and the additive subgroup $G = \langle \binom{1}{1} \rangle = \left\{ \binom{k}{k} : k \in \mathbb{Z} \right\}$. G doesn't arise as a product of two subgroups $S, T \subseteq \mathbb{Z}$, because $1 \in S, T \Rightarrow S, T = \mathbb{Z}$.

Suppose $K \subseteq A \times B$ is a ideal. Let $\pi_1 : A \times B \twoheadrightarrow A$ and $\pi_2 : A \times B \twoheadrightarrow B$ be the standard projection maps. We will first show that $\pi_1(K)$ and $\pi_2(K)$ is a ideal (this is not true for general ring homomorphisms). It suffices to prove this for $\pi_1(K)$. $a_1, a_2 \in \pi_1(K)$, there exists $b_1, b_2 \in B$ such that $(a_1, b_1), (a_2, b_2) \in K$, thus $(a_1 + a_2, b_1 + b_2) \in K \Rightarrow a_1 + a_2 \in \pi_1(K)$. $\forall r \in A$, $(r, 0)(a_1, b_1) =$

$(ra_1, 0) \in K \Rightarrow ra_1 \in \pi_1(K)$. Hence $\pi_i(K)$ is a ideal. Now we will show $K = \pi_1(K) \times \pi_2(K)$. $K \subseteq \pi_1(K) \times \pi_2(K)$ is trivial. $\forall a \in \pi_1(K)$ and $\forall b \in \pi_2(K)$. Note that $(a, 0) \in K$ and $(0, b) \in K$ (multiply by $(1, 0)$ and $(0, 1)$ to any element that starts with a and any element which ends with b). Hence $(a, 0) + (0, b) = (a, b) \in K$. Thus $K = \pi_1(K) \times \pi_2(K)$.

Suppose $\mathfrak{p} \in \operatorname{Spec}(A \times B)$, $\mathfrak{p} = \pi_1(\mathfrak{p}) \times \pi_2(\mathfrak{p})$. Note that

$$(A \times B)/\mathfrak{p} \xrightarrow{\sim} \frac{A}{\pi_1(\mathfrak{p})} \times \frac{B}{\pi_2(\mathfrak{p})}.$$

note that $(A \times B)/\mathfrak{p}$ must be a domain. Assume the contrary that $\pi_1(\mathfrak{p}) \neq A$ and $\pi_2(\mathfrak{p}) \neq B$. Thus

$$(1, 0) * (0, 1) = (0, 0) \in \frac{A}{\pi_1(A)} \times \frac{B}{\pi_2(B)}$$

this contradicts the domain nature of $(A \times B)/\mathfrak{p}$. Thus at least one must be A or B. Thus $\mathfrak{p}$ is of the form $\pi_1(\mathfrak{p}) \times B$ or $A \times \pi_2(\mathfrak{p})$ where $\pi_i(\mathfrak{p})$ is prime. Thus

$$\operatorname{Spec}(A \times B) \coloneqq \operatorname{Spec}(A) \sqcup \operatorname{Spec}(B)$$

□

Problem 6

Let k be a field. Since $R = k[x_1, ..., x_n]$ is a UFD, the ideal (f) is prime for every irreducible polynomial f in R. The following exercise gives some practice in finding irreducible polynomials. (I recommend proving this by hand rather than trying to use big theorems: just analyze what a factorization of the given polynomial would have to look like.)

Show that a polynomial in $k[x_1, ..., x_n]$ of the form $x_n - f(x_1, ..., x_{n-1})$ is irreducible over k. Show that a polynomial of the form $x_n^2 - f(x_1, ..., x_{n-1})$ is irreducible over k if and only if f is not a square in $k[x_1, ..., x_{n-1}]$.

Solution. Note the following that

$$k[x_1, ..., x_n][y] \cong k[x_1, ..., x_n, y]$$

hence if a polynomial in $k[x_1, ..., x_n][y]$ is irreducible over $k[x_1, ..., x_n]$ it is irreducible over k as a element of $k[x_1, ..., x_n]$. We will use this fact to prove the first statement. Suppose that $y - f(x_1, ..., x_n) = AB$ where $y - f(x_1, ..., x_n)$, A, and B are viewed as elements of $k[x_1, ..., x_n][y]$. Either A or (exclusive) B must have degree 0 and the other must have degree 1 (in y). Without loss of generality, A has degree 1 and B has degree 0. Since the coefficient in front of y, in $y - f(x_1, ..., x_n)$, is nonzero hence the constant term of B must be nonzero, hence unit since k is a field. Hence B is a unit in $k[x_1, ..., x_n]$. Hence $y - f(x_1, ..., x_n)$ is irreducible as a element of $k[x_1, ..., x_n]$. Thus this implies that it is irreducible as a element of $k[x_1, ..., x_n, y]$.

Suppose $y^2 - f(x_1, ..., x_n)$ is irreducible. Assume the contrary that f is a square in $k[x_1, ..., x_n]$. Then $\exists g \in k[x_1, ..., x_n]$ such that $g^2 = f$. Then

$$(y - g)(y + g) = y^2 - f.$$

Both $y-g$ and $y+g$ are nonunit, hence y^2-f cannot be irreducible.

Suppose f is not a square. Suppose

$$y^2 - f = A \cdot B$$

where $A, B \in k[x_1, ..., x_n][y]$. WLOG $\deg(A) \geq \deg(B)$.

1. $\deg(A) = 2$: Then B must have a nonzero constant term, hence it must be unit and y^2-f is irreducible.
2. $\deg(A) = 1 \Rightarrow \deg(B) = 1$. Thus $A = g_1 y + h_1$ and $B = g_2 y + h_2$ where $g_1, g_2, h_1, h_2 \in k[x_1, ..., x_n]$. Thus

$$AB = (g_1 y + h_1)(g_2 y + h_2) = g_1 g_2 y^2 + g_1 y h_1 + g_2 y h_2 + h_1 h_2.$$

   $g_1 g_2 = 1 \Rightarrow g_1$ and g_2 are units in $k[x_1, ..., x_n] \Rightarrow g_1$ and g_2 are units in k because the coefficients must be nilpotent, hence 0. Thus $g_1, g_2 \in k^\times$.

$$(g_1 h_2 + g_2 h_1) y = 0 \Rightarrow g_1 h_2 = -(g_1)^{-1} h_1 \Rightarrow g_1^2 h_2 = -h_1$$

Thus

$$\begin{aligned} -f &= h_1 h_2 \\ &= (-g_1^2 h_2) h_2 \\ &= -(g_1 h_2)^2 \Rightarrow f = (g_1 h_2)^2 \end{aligned}$$

hence f is a square and a contradiction occurs. Thus y^2-f is irreducible.